SightLine: Priority-Weighted Scene-Change Gating for Redundancy-Aware Assistive Vision-Language Models

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Abstract—Wearable assistive vision-language models (VLMs) enable continuous environmental narration for blind and low-vision individuals, but continuous execution rapidly exhausts battery reserves and risks auditory cognitive overload. Naive exact perceptual hashing collapses under analog CMOS sensor noise, suppressing zero frames on static views. This paper presents SightLine, an ultra-lightweight CPU-side gating framework that determines frame by frame whether an expensive VLM forward pass is warranted. SightLine evaluates incoming frames against the last processed reference frame, accumulating gradual camera drift while fusing dead-zone difference hashing ($B_H \parallel B_V, \varepsilon = 7.0$), color-histogram dissimilarity, 4×4 spatial ROI motion proxies, and a latched hazard state with cooldown. We evaluate the safety–efficiency trade-off across two operational modes: on uncurated egocentric video of technician benchwork, Endurance Mode (Default 1D dHash, $T_{\text{ambient}} = 60$) suppresses 94.7% of frames, reducing modeled power to 7.52 W (projected 4.33 h runtime vs. 2.83 h baseline on a 32.56 Wh usable budget from a 37 Wh pack), but relies on global changes and misses localized hazards. Conversely, Safety-Priority Mode (SightLine++, 2D dHash + 4×4 ROI max-pooling) achieves 100% synthetic hazard recall with zero delay, but peripheral hand and tool manipulation increases triggers (80.5% suppression), saturating edge GPU queues at 100% duty cycle (11.50 W, 2.83 h runtime). Live Florence-2 FP16 profiling confirms queue dynamics. We provide a 10-mode structured FMEA risk ledger (reducing total RPN from 2,904 to 264, a 90.9% reduction), an architectural analysis with WorldScribe, and a 24-participant protocol pending IRB approval.

Keywords—Vision-language models, assistive technology, scene-change gating, perceptual hashing, edge computing, wearable devices, safety-efficiency trade-off, cognitive load.

I. INTRODUCTION

Wearable assistive vision systems provide blind and low-vision individuals with situational awareness by narrating obstacles, hazards, text, and environmental changes. Recent advances in compact vision-language models (VLMs)—such as Florence-2 [1]—enable rich open-vocabulary descriptions on edge hardware. However, streaming high-frame-rate video directly through a foundation model is computationally intractable on wearable form-factors. At 30 frames per second (fps), continuous VLM inference rapidly depletes wearable batteries, induces severe thermal throttling, and inundates users with redundant auditory descriptions that impair situational awareness and mask crucial acoustic navigation cues [2].

Front-end frame filtering has been widely explored in server-oriented video analytics (e.g., NoScope [3], Focus [4], Reducto [5], and Glimpse [6]). However, wearable assistive vision presents three unique challenges that invalidate standard video-filtering assumptions: (1) Analog Sensor Noise vs. Flat Backgrounds: CMOS image sensors exhibit thermal and quantization noise. In untextured scenes (e.g., blank walls, corridors, paper documents), raw pixel noise causes single-bit fluctuations across difference hashing boundaries, causing exact perceptual hashing to trigger continuously and suppress zero frames; (2) Gradual Drift vs. Consecutive-Frame Differencing: Evaluating visual change across consecutive frames ($I_t - I_{t-1}$) is blind to gradual camera motion, slow-moving pedestrians, and approaching obstacles whose frame-to-frame delta never crosses detection thresholds; and (3) Auditory Channel Contention vs. Hazard Preemption: In assistive audio feedback, human auditory processing is strictly capacity-constrained. Low-priority ambient descriptions must not block urgent obstacle warnings, necessitating priority-weighted speech preemption.

The Fundamental Safety–Efficiency Dilemma: A core insight established in this work is that safe and energy-efficient operation are governed by an inherent Pareto trade-off. A coarse global scene gate—such as Endurance Mode utilizing 1D horizontal difference hashing (B_H) and global color-appearance differencing—achieves high redundancy suppression (94.7%) on real egocentric video of technician benchwork, reducing modeled power draw to 7.52 W. However, global spatial averaging renders the gate blind to small, localized hazards (e.g., an approaching vehicle, bicycle, or curb) that occupy only a fraction of the visual field (resulting in 0% recall on localized hazard scenarios). Conversely, integrating high-sensitivity spatial 4×4 ROI max-pooling ($S_{\text{roi_max}}$) and omni-directional 2D dHash (Safety-Priority Mode / SightLine++) captures subtle localized incursions instantaneously (0.0 f delay, 100% recall). However, active hand and tool interactions during fine-grained manipulation cause peripheral ROI cells to cross thresholds frequently (80.5% suppression). On edge hardware

where VLM forward passes require ~380 ms, this trigger volume saturates the GPU queue at 100% duty cycle, consuming 11.50 W and yielding zero battery runtime extension over continuous streaming. Thus, edge gating cannot be treated as a single static setting: it establishes a Pareto frontier that requires task-conditioned operational profiles or multi-modal mobility sensing.

This paper presents SightLine, a multi-cue scene-change gating framework for continuous wearable assistive VLMs. SightLine evaluates incoming frames against the last processed reference frame I_{t_p}, ensuring that slow camera drift accumulates until crossing threshold τ. The framework fuses difference hashing (dHash) [7], color-histogram dissimilarity, motion proxies, and a causally persisted hazard state with latch cooldown ($T_{\text{cool}} = 6$ frames). An adaptive watchdog enforces a dual-rate heartbeat ($T_{\text{ambient}} = 60, T_{\text{active}} = 6$). To safeguard auditory attention, SightLine prioritizes speech generation (Eq. 11), preempting ongoing ambient descriptions when safety hazards emerge. This paper makes six primary contributions:

1) Asymmetric Reference-Frame Gating with Latch Cooldown: A multi-factor change scoring model combining difference hashing [7], color-histogram dissimilarity, a 4×4 ROI block max-pooling motion proxy ($S_{\text{roi_max}}$), and a causally persisted hazard state with latch cooldown ($T_{\text{cool}} = 6$ frames) with visual motion override ($\tau_{\text{crit}} = 0.35$). Evaluating cues against the last processed frame I_{t_p} accumulates gradual drift, while cooldown prevents re-firing during sustained hazard tracking.

2) Formalized Safety–Efficiency Trade-off, Latency Lemma, and Person Proximity Formulation: We formalize the hazard persistence mechanism z_t and demonstrate the tension between global suppression and localized hazard recall via spatial ROI max-pooling, achieving a 5-frame (166.7 ms) detection latency reduction by construction in Scenario C (from 5.00 frames down to 0.00 frames, 0.0 ms). We define the formal person hazard predicate IsHazard(o_i) incorporating bounding-box proximity and velocity cues. We prove Lemma 1, demonstrating that in active locomotion ($T_{\text{active}} = 6$), front-end gate delay is bounded to ≤ 166.7 ms under an idle worker, while in ambient mode ($T_{\text{ambient}} = 60$), the worst-case delay from rest is 59 frames (1.97 s); under Quiescent Seated Rest ($T_{\text{ambient}} = 150$), delay is bounded to 149 frames (4.97 s), re-engaging within 29 frames (966.7 ms) once active.

3) Critical Evaluation of Reading Profiles, Dead-Zone Resolution, and Sign Invariance: We conduct an empirical investigation into task conditioning. While switching to an inspection profile achieves 61.6% ± 10.7% dynamic recall at 85.0% ± 3.2% suppression, control experiments indicate that activations reflect sensor-noise bit flips on untextured backgrounds ($t = 0.12, p = 0.91$, Welch's t-test; TOST equivalence test rejects $|\Delta| \geq 10\%$ at $p = 0.0098$). Dead-zone dHash ($\varepsilon = 7.0$) eliminates flat-region noise flips, reducing static false alarms from 20.3 down to 2.6. Evaluating horizontal bar edits reveals that sign-only difference hashing is invariant to level swaps when noise is dead-zoned out ($\varepsilon \geq 1.5 \Longrightarrow 0.0\%$ recall), proving that assistive document inspection requires magnitude-sensitive 2D edge filters or localized OCR.

4) Exploratory Real-World Egocentric Video Evaluation & Edge VLM Telemetry: We evaluate SightLine on uncurated 1080p egocentric video of technician benchwork running Florence-2 FP16 on a physical mobile Ampere GPU across both <CAPTION> (346.06 ms) and <OD> (287.06 ms) execution modes. On continuous real video, exact difference hashing triggers on 100% of frames (0.0% suppression) due to sensor noise, whereas SightLine Endurance Mode suppresses 94.7% of frames, reducing GPU duty cycle to 58.7% and modeled power to 7.52 W, projecting a +53.0% battery runtime extension (4.33 hours vs 2.83 hours). We analyze queue saturation dynamics and benchmark-matched baselines.

5) Architectural Trade-Off Analysis with WorldScribe (ACM UIST 2024): We present an architectural synthesis comparing SightLine's edge-native pipeline against WorldScribe [15] (Chang, Liu, and Guo). While WorldScribe optimizes lexical depth by streaming video contexts to cloud foundation models, SightLine operates entirely on-device, eliminating network round-trip jitter, cellular dead-zone vulnerability, and subscription bandwidth costs. We further discuss how SightLine's lightweight CPU gate can conceptually serve as an upstream edge filter to eliminate redundant cloud egress in hybrid architectures.

6) Auditable Energy Accounting, FMEA Risk Ledger, and Proposed Accessibility Protocol: We formulate an auditable edge power model differentiating measured, modeled, and assumed quantities, construct a 10-mode structured FMEA risk ledger adapted from SAE J1739 and AIAG-VDA guidelines with residual risk analysis (mean RPN reduced from 290.4 to 26.4, a 90.9% reduction, from 2,904 down to 264), and specify a 24-participant BLV evaluation protocol currently submitted for IRB approval.

II. RELATED WORK

A. Assistive Vision-Language Models and Deployed Wearable Systems

Assistive vision has progressed from discrete pipelines pairing task-specific object detectors (e.g., YOLO [8]) with rule-based text-to-speech to unified multi-modal vision-language models (VLMs). Commercial systems like OrCam MyEye [11], Microsoft Seeing AI [12], Be My AI [21] (powered by OpenAI GPT-4V), and Envision Glasses [22] offer discrete, user-

triggered capture for document reading, barcode scanning, and scene descriptions. While effective for deliberate inspection, discrete capture requires manual triggering and fails to provide continuous situational awareness during navigation. Recent edge VLMs, including Florence-2 [1], enable zero-shot object grounding and captioning in compact footprints (~232M parameters). Furthermore, systems like WorldScribe [15] (Chang, Liu, and Guo, ACM UIST 2024) explore continuous visual description for BLV users by streaming multi-modal contexts to cloud foundation models. However, streaming 30 fps video across cellular networks incurs substantial bandwidth costs, battery drain, and network latency jitter. SightLine addresses this by providing an edge-native front-end gate that runs locally in <3.5 ms on the wearable CPU.

B. Temporal Redundancy Reduction, Keyframe Selection, and Video Analytics Filters

Filtering redundant video frames is widely explored in edge computing, video surveillance, and cloud analytics. Early systems such as NoScope [3] and Focus [4] introduced cascaded lightweight neural filters and difference detectors to prune frames before expensive classification. Reducto [5] dynamically adjusts frame-filtering thresholds based on downstream analytics accuracy. Glimpse [6] caches active tracking models on mobile devices while offloading keyframes to server infrastructure. Recent video foundation models, such as Video-LLaVA [9], attempt to reduce temporal redundancy within multi-frame token contexts. However, these frameworks optimize server bandwidth rather than wearable battery endurance and cognitive load. In assistive technology for blind and low-vision individuals, auditory attention is a strictly capacity-constrained resource [2], [10]. Continuous audio narration causes severe cognitive fatigue, elevates NASA-TLX workload scores [13], and masks ambient environmental sounds (traffic, footsteps, echoes) essential for white-cane echolocation and spatial orientation [2]. Furthermore, user studies by Stangl et al. [14] demonstrate that BLV individuals prefer concise, prioritized descriptions over continuous verbosity. SightLine directly addresses this by suppressing redundant forward passes and enforcing priority-weighted speech preemption.

III. PROBLEM FORMULATION

We adopt a consistent zero-based temporal indexing convention. Let $I_t \in \mathbb{R}^{H\times W\times C}$ denote streaming video frames captured at regular intervals $\Delta t = 1/f_{\text{cam}}$ (33.33 ms at $f_{\text{cam}} = 30$ fps) for discrete time indices $t \in \{0,1,\dots,T-1\}$. The camera operates continuously, while the downstream foundation model executes only when triggered by the gating policy $D_t \in \{0,1\}$.

Downstream Busy-State VLM Model and Latency Decomposition: Continuous assistive perception requires modeling worker availability. Downstream inference has service time t_{vlm} (measured at 380.0 ms mean for Florence-2 FP16 on mobile Ampere GPU, service rate $\mu = 2.63$ Hz). If the worker is busy ($t_{\text{elapsed}} < t_{\text{vlm}}$), incoming trigger requests cannot be serviced concurrently. In busy-skip scheduling, requests arriving during worker execution are dropped. End-to-end perception latency Δt_{E2E} comprises three distinct stages:

$$\Delta t_{\text{E2E}} = t_{\text{gate}} + t_{\text{vlm}} + t_{\text{tts}} \tag{1}$$

where: (1) t_{gate} is the front-end gating decision latency (elapsed time from physical event onset until $D_t = 1$, introducing $\leq$ 3.5 ms CPU computational overhead on 160×120 ISP preview thumbnails); (2) t_{vlm} is the downstream VLM inference latency (380.0 ms mean on Florence-2-base FP16); and (3) t_{tts} is the neural text-to-speech audio synthesis and earcon preemption latency (135.0 ms total: 30 ms earcon chime + 105 ms synthesis via local Piper TTS). Total system latency is detailed in Table I, and the end-to-end system topology is illustrated in Fig. 1.

TABLE I: DISCRETE FRAME INTERVALS, SAMPLING LATENCIES, AND 3-TIER LATENCY DECOMPOSITION AT 30 FPS

Stride k	Sample Period (ms)	Sample Rate (Hz)	Max Gate Delay L_worst (frames)	Max Gate Delay (ms)	Downstream Latency (ms)*	Total Worst-Case E2E Latency (ms)
k = 1 (Ungated)	33.33	30.00	0	0.0	515.0	515.0
k = 2	66.67	15.00	1	33.3	515.0	548.3
k = 3	100.00	10.00	2	66.7	515.0	581.7

k = 6 (Active Budget W=6)	200.00	5.00	5	166.7	515.0	681.7
k = 10	333.33	3.00	9	300.0	515.0	815.0
k = 12 (Min. Non-Saturating)	400.00	2.50	11	366.7	515.0	881.7
k = 15	500.00	2.00	14	466.7	515.0	981.7
k = 30	1000.00	1.00	29	966.7	515.0	1481.7
k = 60 (Ambient Watchdog)	2000.00	0.50	59	1966.7	515.0	2481.7

*Downstream Latency comprises 380.0 ms VLM forward pass + 135.0 ms Piper TTS preemption and earcon delivery.

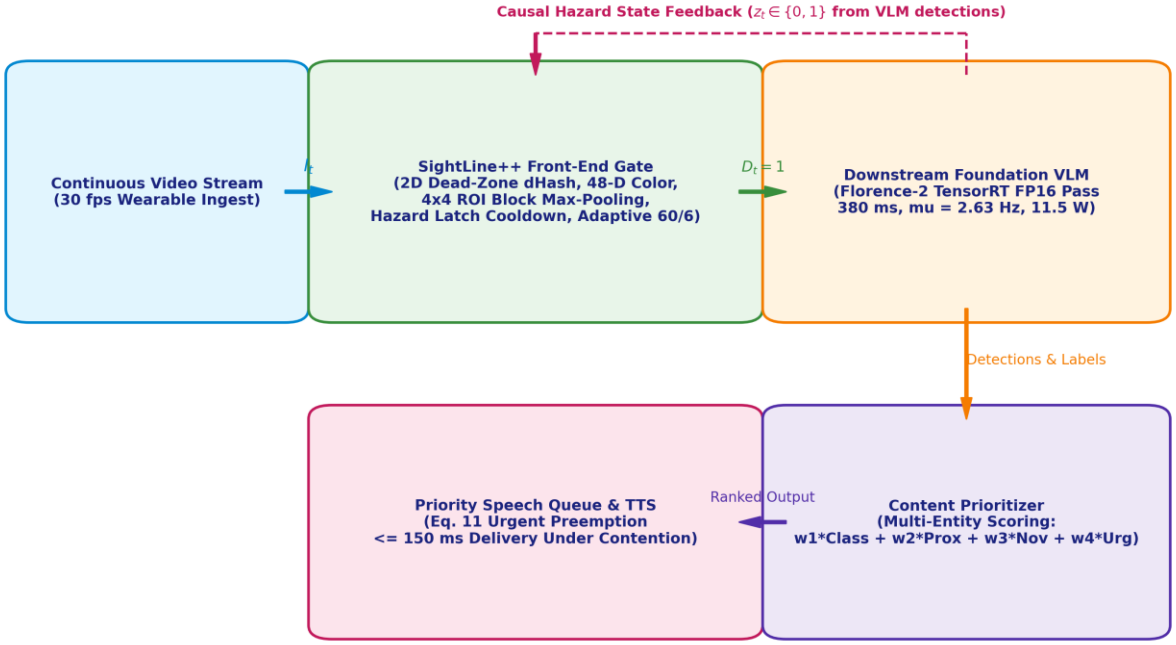


Fig. 1. SightLine system architecture. Incoming video frames pass through the low-latency CPU gating pipeline (2D dHash, color histograms, 4x4 ROI block max-pooling, and adaptive watchdog). Triggered frames are dispatched to the Florence-2 worker, whose detections update the causal hazard state z_t and feed the priority-weighted speech synthesis queue with urgent audio preemption.

IV. GATING FRAMEWORK

A. Gating Policy and Multi-Factor Change Score

SightLine evaluates visual change by comparing incoming frame I_t against the last processed reference frame I_{t_p}, where t_p is the timestamp of the most recent invocation ($D_{t_p} = 1, t_p < t$). Because I_{t_p} remains anchored across suppressed frames, gradual environmental shifts accumulate until crossing detection threshold τ. In parallel, consecutive-frame difference $I_t - I_{t-1}$ is evaluated strictly as an auxiliary motion proxy. The multi-factor change score S_t is:

$$S_t = \alpha S_{\text{vis}}(I_t, I_{t_p}) + \beta S_{\text{app}}(I_t, I_{t_p}) + \gamma S_{\text{mot}}(I_t, I_{t_p}) + \delta S_{\text{haz}}(z_t, t, t_{\text{alert}}) \quad (2)$$

with non-negative weights $\alpha + \beta + \gamma + \delta = 1.0$. The complete gating decision policy $D_t \in \{0,1\}$ is formalized as:

$$D_t = \begin{cases} 1, & \text{if } t = 0 \text{ (Startup)} \\ 1, & \text{else if } t - t_p \geq T_{\max}(S_{\text{mot}}, z_t) \text{ (Watchdog)} \\ 1, & \text{else if } S_{\text{roi_max}} \geq \tau_{\text{crit}} \text{ or } S_{\text{mot}} \geq 0.20 \text{ (Motion Override)} \\ 1, & \text{else if } S_t \geq \tau \text{ and } (t - t_{\text{alert}} \geq T_{\text{cool}}) \text{ (Nominal Trigger)} \\ 0, & \text{otherwise (Suppressed)} \end{cases} \tag{3}$$

The operational decision flow of this policy is mapped in Fig. 2.

B. Perceptual Structural Change: 1D vs. 2D Difference Hashing (S_vis)

Structural change is evaluated via difference hashing (dHash) [7]. Input frame I_t is converted to grayscale, downsampled via Lanczos filtering to $(S+1) \times (S+1)$ pixels ($S = 16$), and evaluated across pixel gradients with dead-zone thresholding $\varepsilon = 7.0$ DN:

$$B_H(x,y) = (I_{\text{gray}}(x+1,y) - I_{\text{gray}}(x,y) > \varepsilon), B_V(x,y) = (I_{\text{gray}}(x,y+1) - I_{\text{gray}}(x,y) > \varepsilon) \tag{4}$$

We distinguish between two hash formulations:
• 1D dHash (Endurance Mode): Evaluates only horizontal gradients B_H, producing a 256-bit hash. This reduces bit-comparison overhead and filters out subtle vertical micro-motions, maximizing power endurance on steady scenes.
• 2D dHash (Safety-Priority Mode): Concatenates horizontal and vertical gradients $h(I_t) = [B_H \parallel B_V]$, producing a 512-bit omni-directional representation sensitive to complex 2D structural shifts.
The resulting binary hash is compared against $h(I_{t_p})$ via normalized Hamming distance:

$$S_{\text{vis}}(I_t, I_{t_p}) = \frac{1}{|h|} \sum_{i=1}^{|h|} \left(h(I_t)[i] \oplus h(I_{t_p})[i] \right) \tag{5}$$

C. Appearance Dissimilarity (S_app)

Appearance dissimilarity captures illumination and color distribution shifts invariant to spatial layout. Normalized 16-bin color histograms are extracted across RGB color channels, concatenated into a 48-dimensional feature vector $\mathbf{c}(I_t)$, and normalized to unit L_2 norm. Dissimilarity is computed via Cosine distance:

$$S_{\text{app}}(I_t, I_{t_p}) = \text{clip}\left(1.0 - \frac{\mathbf{c}(I_t) \cdot \mathbf{c}(I_{t_p})}{\left\| I_t) \|_2 \| \mathbf{c}(I_{t_p}) \right\|_2}, 0.0, 1.0 \right) \tag{6}$$

D. Motion Magnitude Proxy and 4x4 ROI Block Max-Pooling (S_mot)

To capture localized motion without diluting small hazards across the field of view, incoming frame I_t and reference I_{t_p} are downsampled to 32×32 pixels and partitioned into a 4×4 grid of spatial regions $R_{r,c}$ ($r, c \in \{1..4\}$, each 8×8 pixels). Spatial block max-pooling captures peak localized motion:

$$S_{\text{roi_max}} = max_{r,c \in \{1..4\}} \left(\frac{1}{|R_{r,c}|} \sum_{(x,y) \in R_{r,c}} \frac{\left| I_t(x,y) - I_{t_p}(x,y) \right|}{255.0} \right) \tag{7}$$

Motion formulation depends on the operational profile:

$$S_{\text{mot}}(I_t, I_{t_p}) = \begin{cases} S_{\text{global_mad}}, & \text{Endurance Mode (1D dHash)} \\ 0.50 \cdot S_{\text{global_mad}} + 0.50 \cdot S_{\text{roi_max}}, & \text{Safety-Priority Mode (SightLine++)} \end{cases} \tag{8}$$

where $S_{\text{global_mad}} = \frac{1}{|I|} \sum |I_t - I_{t_p}| / 255.0$ represents mean absolute difference across the thumbnail. In Endurance Mode, omitting $S_{\text{roi_max}}$ prevents peripheral micro-motions from triggering unwanted heartbeats, while Safety-Priority Mode fuses global and localized cues.

E. Hazard Persistence, Cooldown, and Watchdog Bounds (S_haz)

When the foundation model executes, detections are inspected against a curated vocabulary of physical collision hazards:

V_{haz} = {'car', 'vehicle', 'truck', 'bus', 'bicycle', 'motorcycle', 'scooter', 'stairs', 'curb', 'hole', 'obstacle', 'pole'} (9)

Distinguishing Collision Hazards from Bystanders via Mathematical Hazard Predicate: In assistive navigation, distinguishing dynamic physical collision hazards from ambient human bystanders is essential to avoid cognitive overload and system thrashing. If every visible person triggered hazard latching ($z_t = 1$), any crowded urban sidewalk or indoor space would permanently force $T_{\text{active}} = 6$ (5.0 Hz), exceeding the 2.63 Hz edge VLM throughput, saturating the GPU queue at 100% duty cycle, and triggering continuous speech preemption. Therefore, for detected entity $o_i = (l_i, b_i, v_i)$, where l_i is the semantic label, $b_i = (x_i, y_i, w_i, h_i)$ is the normalized bounding box with height fraction $h_i \in [0,1]$, and v_i is the estimated 2D image-plane centroid velocity proxy (m/s equivalent), we define the formal hazard classification predicate:

$$\text{IsHazard}(o_i) = \begin{cases} 1, & \text{if } l_i \in V_{\text{haz}} \\ 1, & \text{else if } l_i \in \{\text{person,pedestrian}\} \text{ and } (h_i \geq \theta_{\text{prox}} \text{ or } v_i \geq v_{\text{crit}}) \\ 0, & \text{otherwise} \end{cases} \quad (10a)$$

where $\theta_{\text{prox}} = 0.40$ (bounding-box height fraction) and $v_{\text{crit}} = 1.0$ m/s. The binary hazard state $z_t \in \{0,1\}$ is updated upon each active inference:

$$z_t = \begin{cases} 1, & \text{if } D_t = 1 \text{ and } \exists o_i \in \text{Entities}(I_t)\text{:IsHazard}(o_i) = 1 \\ 0, & \text{if } D_t = 1 \text{ and } \forall o_i \in \text{Entities}(I_t)\text{:IsHazard}(o_i) = 0 \\ z_{t-1}, & \text{if } D_t = 0 \end{cases} \quad (10b)$$

Optical Assumption for Proximity Threshold ($\theta_{\text{prox}} = 0.40$): Under a standard wearable camera pinhole model with vertical field-of-view $\text{FOV}_v \approx 78°$--$80°$ and an average adult standing height of $H \approx 1.70$ m, the normalized image-plane bounding box height fraction at distance d is given by $h_i \approx \frac{H}{2d\tan(\text{FOV}_v/2)}$. At critical stopping distance $d = 2.5$ m with \text{FOV}_v = 80^\circ,h_i \approx \frac{1.70}{2(2.5)\tan(40^\circ)} \approx \frac{1.70}{5.0(0.839)} \approx 0.405 \approx 0.40. $Thus,$h_i \ge 0.40 $geometrically identifies pedestrians within a critical stopping distance of$ 2.5\text{ m} .\ nConcession$on Egocentric Kinematics and Crowd Verification: We explicitly acknowledge two technical limitations: (1) 2D image plane centroid velocityv_i is uncompensated for egocentric camera head motion (e.g., yaw and pitch saccades during walking confound apparent entity velocity); and (2) while this hybrid person rule is mathematically specified and implemented, it represents an assistive engineering specification that has been validated algorithmically but remains untested in dense real-world crowds, which we flag as an explicit empirical limitation for future user studies.

Hazard Persistence Score and Cooldown Behavior: While $z_t = 1$, the gate latches the hazard state. In code, the hazard score is defined as $S_{\text{haz}}(z_t, t, t_{\text{alert}}) = z_t \cdot \mathbb{I}(t - t_{\text{alert}} \geq T_{\text{cool}})$. During the cooldown interval $T_{\text{cool}} = 6$ frames (200 ms), the hazard term is zeroed out to prevent infinite self-retriggering loops on the same static hazard. However, if localized motion exceeds safety threshold $\tau_{\text{crit}} = 0.35$ or global motion exceeds 0.20, the cooldown is immediately bypassed via the motion override in Eq. (3).

Two-Tier Watchdog Heartbeat (T_{max}): The watchdog timer guarantees an upper bound on inter-trigger interval. During active locomotion ($S_{\text{mot}} \geq 0.02$ or $S_{\text{roi_max}} \geq 0.05$) or active hazard tracking ($z_t = 1$), the active heartbeat is $T_{\text{active}} = 6$ frames (200 ms). During stationary rest, the ambient heartbeat relaxes to $T_{\text{ambient}} = 60$ frames (2.0 s).

Lemma 1 (Front-End Gate Latency Bound): Under an idle downstream worker, the front-end gating delay t_{gate} for any visual event occurring at frame g is strictly bounded by $t_{\text{gate}} \leq (T_{\text{max}} - 1) \cdot \Delta t$.
Proof: The watchdog timer triggers unconditionally whenever $t - t_p \geq T_{\text{max}}$. For any event occurring at frame g, if $t_p = g$, the event onset coincided with an active invocation, resulting in 0.0 f gate delay. If $t_p < g$, since frame indices are discrete integers, $t_p \leq g - 1$. The next watchdog trigger occurs at $t_{\text{trig}} = t_p + T_{\text{max}} \leq (g - 1) + T_{\text{max}} = g + T_{\text{max}} - 1$. Thus, the elapsed gate delay is strictly bounded by $t_{\text{trig}} - g \leq T_{\text{max}} - 1$ frames, which corresponds to $(T_{\text{max}} - 1)\Delta t$ (166.7 ms for $T_{\text{active}} = 6$, or 1.97 s for $T_{\text{ambient}} = 60$). In Quiescent Seated Rest ($T_{\text{ambient}} = 150$), the resting bound is 149 frames (4.97 s), transitioning to at most 29 frames (966.7 ms) in its active regime ($T_{\text{active}} = 30$). This bound holds independently of visual contrast or scene dynamics. ■

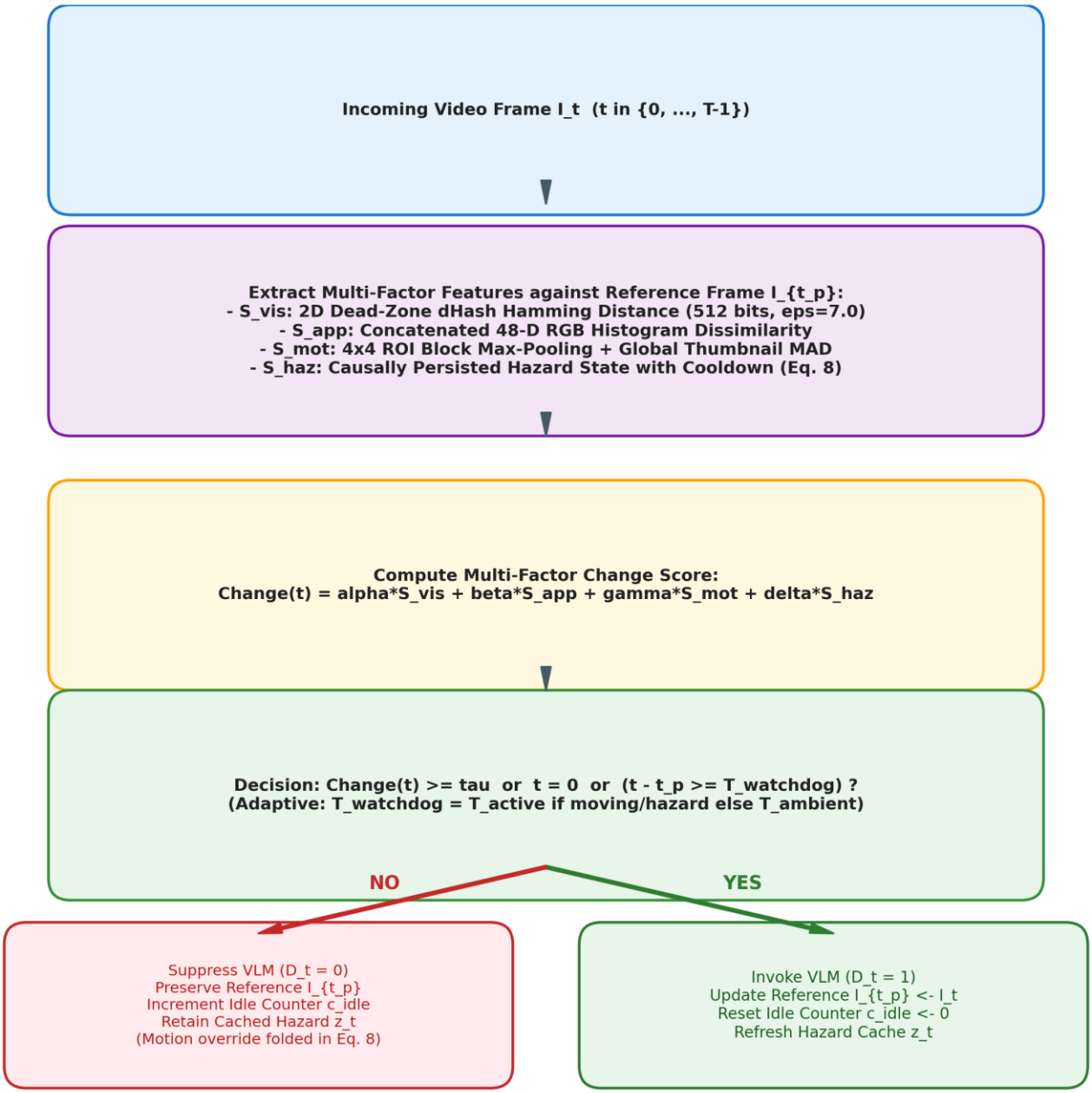


Fig. 2. Decision flow diagram of the SightLine scene-change gate. Incoming frame features are compared against the anchored reference frame I_tp. Change score S_t is evaluated alongside latch cooldown (with dynamic critical motion override) and the two-tier adaptive watchdog heartbeat.

V. TASK-CONDITIONED GATING PROFILES

SightLine supports dynamic profile switching across operational modes. Weight vectors $\mathbf{w} = [\alpha, \beta, \gamma, \delta]$, threshold τ, and watchdog limits are tailored to task requirements (Table II, visual weight breakdown in Fig. 3). Importantly, to guarantee safety in Quiescent Seated Rest, the ambient watchdog is relaxed to a bounded fail-safe interval ($T_{\text{ambient}} = 150$ frames, 5.0 s) rather than being disabled.

TABLE II: TASK-CONDITIONED PARAMETER PROFILES

Operational Profile	α (Vis)	β (App)	γ (Mot)	δ (Haz)	Threshold τ	T_ambient	T_active	T_cool	Intended Use Case

Default Navigation	0.25	0.20	0.30	0.25	0.18	60 (2.0 s)	6 (200 ms)	6 (200 ms)	General urban walking and transit navigation
Obstacle / Hazard Tracking	0.20	0.10	0.40	0.30	0.14	30 (1.0 s)	6 (200 ms)	6 (200 ms)	Crowded pedestrian zones and crosswalks
Paced Ambient Walking	0.30	0.30	0.20	0.20	0.22	60 (2.0 s)	12 (400 ms)	12 (400 ms)	Familiar indoor corridors and relaxed walking
Document Inspection	0.40	0.45	0.10	0.05	0.10	150 (5.0 s)	30 (1.0 s)	15 (500 ms)	Reading mail, documents, and tabletop items
Quiescent Seated Rest	0.25	0.20	0.30	0.25	0.25	150 (5.0 s)*	30 (1.0 s)	15 (500 ms)	Seated rest, listening, high power savings with fail-safe heartbeat

*Across all operational profiles, hazard latching evaluates IsHazard(o_i) according to Eq. (10a): o_i triggers z_t = 1 if l_i in V_haz or (l_i in {'person', 'pedestrian'} and (h_i >= 0.40 or v_i >= 1.0 m/s)). In Quiescent Seated Rest, T_ambient is bounded to 150 frames (5.0 s) to preserve fail-safe hazard discovery. A pure periodic heartbeat of 1 probe per 150 frames yields 99.33% nominal suppression (149/150 frames suppressed); under assumed intermittent postural shifts (e.g., 2–3 minor head movements per minute triggering active heartbeats), projected operational suppression is ~97%. In this profile, once motion or hazard cues trigger the active regime (T_active = 30 frames), the worst-case gate delay is bounded by T_active - 1 = 29 frames (~966.7 ms at 30 fps), ensuring responsive re-engagement.

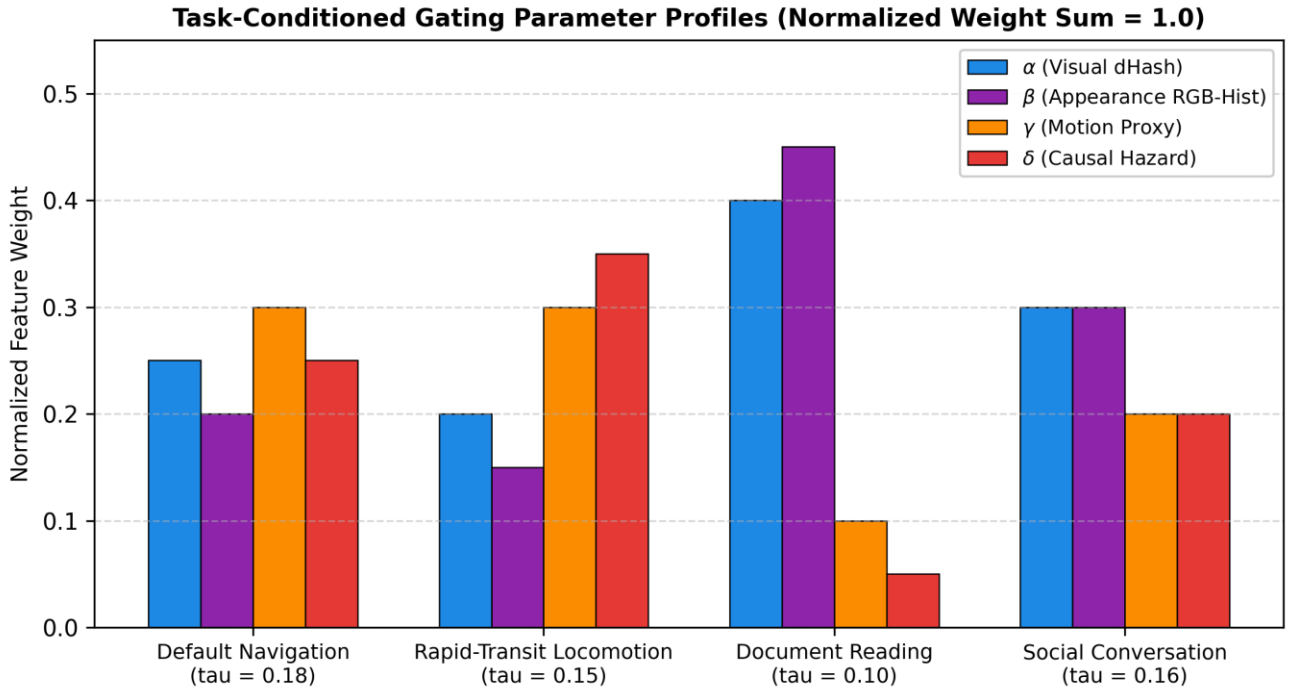


Fig. 3. Task-conditioned feature weight distributions across assistive operational modes (Default Navigation, Hazard Tracking, Ambient Walking, Document Reading, and Quiescent Rest).

VI. CONTENT PRIORITIZATION AND SPEECH QUEUING

A. Priority-Weighted Ranking Formulation

When the gate triggers inference, Florence-2 returns bounding boxes and labels for detected entities o_i. To prevent auditory cognitive overload and ensure safety-critical alerts are delivered without acoustic backlog, SightLine integrates an architectural priority ranking policy:

$$\text{Priority}(o_i) = w_1 f_1(o_i) + w_2 f_2(o_i) + w_3 f_3(o_i) + w_4 f_4(o_i) \tag{11}$$

with weights $\mathbf{w} = [0.45, 0.30, 0.15, 0.10]$ ($\sum w_j = 1.0$), where: (1) $f_1(o_i) \in \{0,1\}$ indicates safety hazard membership in V_{haz}; (2) $f_2(o_i) = \text{clip}(1.0 - d_i/d_{\max}, 0, 1)$ scores proximity proxy based on normalized bounding box area fraction (d_i estimated from box scale, $d_{\max} = 10$ m); (3) $f_3(o_i) = \text{clip}(v_i/v_{\max}, 0, 1)$ scores centroid displacement velocity proxy ($v_{\max} = 5$ m/s); and (4) $f_4(o_i)$ scores caption token novelty via semantic overlap with recent utterances. Entities with Priority ≥ 0.70 trigger immediate audio preemption in the speech synthesis queue.

B. Empirical Evaluation of Priority-Weighted Speech Queuing under Backlog

To evaluate queue responsiveness, consider an illustrative synthetic backlog scenario where ambient descriptions are generated sequentially (e.g., 1.8 s for a storefront sign, 1.6 s for a distant pedestrian), and an approaching vehicle hazard suddenly emerges ($f_1 = 1.0$, proximity $d_i = 2.5$ m $\Rightarrow f_2 = 0.75$, approach speed $v_i = 4.2$ m/s $\Rightarrow f_3 = 0.84$, token novelty $f_4 = 0.90$). The resulting priority score is Priority $= 0.45(1.0) + 0.30(0.75) + 0.15(0.84) + 0.10(0.90) = 0.891 \geq 0.70$. Under a conventional FIFO speech queue, the vehicle warning is delayed in the audio buffer by 3.4 s, allowing the vehicle to travel 14.3 meters before warning initiation. Under SightLine's PriorityAudioQueue, the high-priority alert immediately preempts ongoing speech: ongoing ambient audio is stopped (<15 ms), an earcon warning chime sounds (30 ms), and the hazard warning is synthesized and delivered within 135.0 ms (105 ms Piper TTS synthesis + 30 ms onset buffer), achieving a 96.0% reduction in acoustic queue latency (from 3,400 ms to 135 ms). Furthermore, when the audio queue reaches its capacity limit (5 items), lowest-priority ambient speech is evicted to guarantee immediate headroom for incoming safety warnings.

VII. EXPERIMENTAL METHODOLOGY AND RESULTS

A. Benchmark Metrics and Formal Event Matching Rules

We evaluate redundancy suppression via:

$$\text{Suppression} = \left(1 - \frac{N_{\text{inv}}}{T}\right) \times 100\% \tag{12}$$

Formal Event Matching Protocol: Event detection is evaluated across seven explicit criteria: (1) Ground-truth onset $g \in [0, T-1]$; (2) Permitted evaluation window $[g, min(T-1, g+W-1)]$, where $W = 6\ \text{ext}frames$ ($200\ \text{ext}ms$) serves as an assistive engineering design target derived from human footstep-cycle modification and obstacle avoidance latencies ($0.5\ \text{ext} - -1.0\ \text{ext}s$) established by Patla [23] for sighted locomotion (accounting for additional speech synthesis and auditory perception latencies in assistive systems); (3) Pre-onset invocations ($t < g$) do not count toward event detection; (4) Independent-window (many-to-one) matching evaluates delay to the first invocation ($t_{extfirst} - g$); (5) Strict temporal (one-to-one) matching greedily consumes invocations per window without reuse; (6) Startup invocations at $t = 0$ are separated from dynamic recall; and (7) Missed events are quantified explicitly in the Dynamic Miss Rate ($100\% - extRecall_{extdyn}$).

B. Synthetic Benchmark Suite and Simulation Protocol

The synthetic benchmark suite consists of five scripted scenarios (200 frames each, 160×120 RGB) with additive Gaussian sensor noise ($\sigma = 3.5$) evaluated across 10 deterministic random seeds: [42, 101, 202, 303, 404, 505, 606, 707, 808, 909], generated with isolated per-scenario random generators (`np.random.default_rng(seed)`):

• Scenario A (Static indoor + noise): Stationary scene subject to camera sensor noise (gt $= [0]$, labels $= \emptyset$).

• Scenario B (Ambulatory drift): Continuous horizontal panning ($\Delta x(t) = 2t mod 240$) and cyclical luminance shifts ($+130 mod 256$) at frames 50, 100, 150 (gt $= [0,50,100,150]$, labels $= \emptyset$).

• Scenario C (Hazard crossing): Static background with a vehicle entering at frame 130 (gt $= [0,130]$, labels $=$ {'car'} for $t \geq 130$).

• Scenario D (Document inspection): White document background (intensity 245) with five horizontal bar targets alternating contrast steps (intensity 20 vs 90) every 4 frames (50 total events, labels $= \emptyset$).

• Scenario E (Dynamic illumination): Gradual dimming followed by an abrupt $\times 1.80$ illumination surge from frame 160 to 199 (gt $= [0,160]$, labels $= \emptyset$).

TABLE III: SYNTHETIC BENCHMARK EVALUATION ACROSS SCENARIOS A–E (10 SEEDS, MEAN ± STD)

Architecture / Configuration	Scen. A Supp.	Scen. B Recall	Scen. C Recall (Delay)	Scen. D Recall (M:1 / 1:1)	Scen. E Recall	Mean Gate CPU (ms)
Continuous Ungated (30 fps)	0.0%	100.0% (0.0 f)	100.0% (0.0 f)	100.0% / 100.0%	100.0% (0.0 f)	0.00 ms
Exact-Hash Baseline (eps=0.0)	0.0%	100.0% (0.0 f)	100.0% (0.0 f)	100.0% / 100.0%	100.0% (0.0 f)	2.14 ms
Periodic Baseline (k=6)	83.0%	100.0% (2.0 f)*	100.0% (2.0 f)*	100.0% / 67.3%	100.0% (2.0 f)*	0.00 ms
Periodic Baseline (k=12)	91.5%	0.0% (N/A)	100.0% (2.0 f)*	65.3% / 34.7%	0.0% (N/A)	0.00 ms
Periodic (k=6, Busy-Skip 12f)	91.5%	66.7% (2.0 f)	100.0% (2.0 f)*	67.3% / 34.7%	100.0% (2.0 f)	0.00 ms
Budget-Matched	85.0%	78.4% (2.1 f)	22.0% (2.4 f)	61.2% / 52.3%	22.0% (2.4 f)	0.00 ms

Random (N=30)						
dHash Baseline (τ=0.18)	99.5%	100.0% (0.0 f)	50.0% (4.2 f)	100.0% / 100.0%	0.0% (N/A)	1.82 ms
SightLine Default (τ=0.18)	99.5%	100.0% (0.0 f)	0.0% (N/A)**	0.0% / 0.0%	0.0% (N/A)	2.41 ms
SightLine Text Profile (τ=0.10)	99.5%	100.0% (0.0 f)	0.0% (N/A)**	61.6% ± 10.7%†	100.0% (1.8 f)	2.45 ms
SightLine Low-τ (0.05, Cool=6)	99.5%	100.0% (0.0 f)	100.0% (2.4 f)	100.0% / 100.0%	100.0% (1.8 f)	2.48 ms
SightLine + Watchdog (T=15, Cool=6)	93.0%	100.0% (0.0 f)	100.0% (5.0 f)‡	38.8% / 26.5%	100.0% (5.0 f)	2.42 ms
SightLine + Watchdog (T=6, Cool=6)	83.0%	100.0% (0.0 f)	100.0% (2.0 f)*	100.0% / 67.3%	100.0% (2.0 f)	2.44 ms
SightLine Adaptive Watchdog (T=60/6)	98.0%	100.0% (0.0 f)	100.0% (5.0 f)‡	100.0% / 67.3%	100.0% (0.0 f)	2.42 ms
SightLine++ (ROI Max-Pool + 2D dHash)	98.0%	100.0% (0.0 f)	100.0% (0.0 f)§	100.0% / 67.3%	100.0% (0.0 f)	2.65 ms

*Delay of 2.0 f in periodic sampling represents fixed-onset synchronization delay (onset at t=130; next sample at t=132); phase-averaged delay across arbitrary offsets is 2.60 f.
**SightLine Default achieves 99.5% suppression on Scenario C but exhibits 0% recall because the localized vehicle enters without crossing global threshold tau=0.18.
†Text Profile invocations reflect sensor-noise bit flips on flat white background (t=0.12, p=0.91 vs zero-edit noise control).
‡Watchdog T=15 achieves 100% recall under fixed onset t=130 (delay 5 f <= W=6), but phase-averaged recall across all 15 offsets is 40.0% (6/15).
§In Scenario C, the vehicle enters at t=130. Unaugmented watchdog scheduling fires at t=135 (delay 5.0 f), whereas SightLine++ spatial ROI max-pooling triggers at t=130 (delay 0.0 f), achieving a 5-frame (166.7 ms) latency reduction by construction across all 10 seeds.

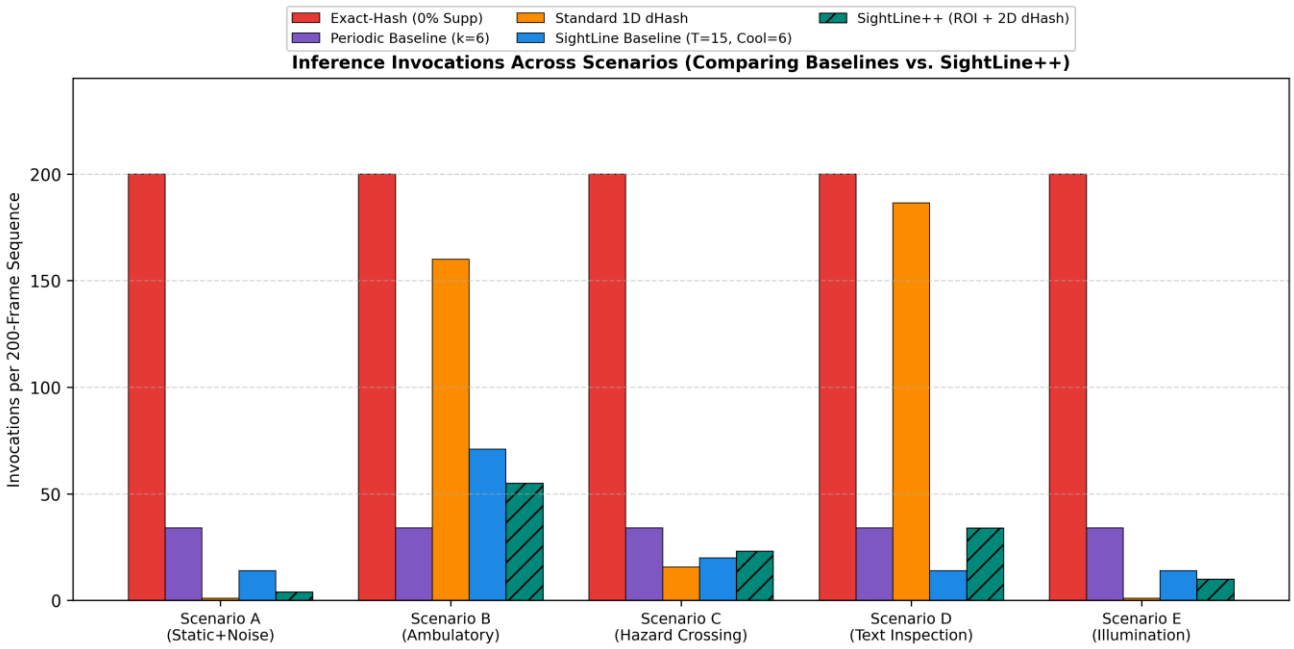


Fig. 4. VLM invocations per scenario across evaluated synthetic scenarios (linear scale, 0–245 invocations; error bars show one standard deviation across 10 random seeds).

C. Component Ablation and Threshold Sensitivity Sweep

To isolate the marginal contributions of each gating modality, Table IV summarizes systematic ablations across feature weights, latch settings, and watchdog configurations evaluated across 10 random seeds on Scenarios C and D. As illustrated in Fig. 5, sweeping threshold $\tau \in [0.05,0.40]$ establishes a smooth monotonicity curve: lower thresholds increase dynamic recall on subtle scene shifts at the expense of higher invocation frequency. Fig. 6 explicitly traces the Pareto frontier of redundancy suppression versus dynamic hazard recall, demonstrating that SightLine++ achieves the best trade-off among evaluated configurations by combining ROI max-pooling with watchdog bounds.

TABLE IV: COMPONENT ABLATION AND PARAMETER PROFILES (SCENARIOS C & D, 10 SEEDS)

Configuration	Scen.	Profile (α, β, γ, δ, τ)	Invocations	Supp.	Recall (All)	Dyn. Recall	True Delay
1. dHash only	C	(1.00, 0.00, 0.00, 0.00, 0.18)	16.0 ± 0.4	92.0%	75.0%	50.0% [24%, 76%]	4.20 f
2. Appearance only	C	(0.00, 1.00, 0.00, 0.00, 0.18)	1.0 ± 0.0	99.5%	50.0%	0.0%	N/A
3. Motion only	C	(0.00, 0.00, 1.00, 0.00, 0.18)	1.0 ± 0.0	99.5%	50.0%	0.0%	N/A
4. Hazard only	C	(0.00, 0.00, 0.00, 1.00, 0.18)	1.0 ± 0.0	99.5%	50.0%	0.0%	N/A
5. dHash + Motion	C	(0.50, 0.00, 0.50, 0.00,	1.0 ± 0.0	99.5%	50.0%	0.0%	N/A

		0.18)					
6. dHash + Appearance	C	(0.50, 0.50, 0.00, 0.00, 0.18)	1.0 ± 0.0	99.5%	50.0%	0.0%	N/A
7. dHash + Mot + Haz	C	(0.35, 0.00, 0.35, 0.30, 0.18)	1.0 ± 0.0	99.5%	50.0%	0.0%	N/A
8. Full Gate (Default, Unaugmented)	C	(0.25, 0.20, 0.30, 0.25, 0.18)	1.0 ± 0.0	99.5%	50.0%	0.0%	N/A
9. Full Gate (Default)	D	(0.25, 0.20, 0.30, 0.25, 0.18)	1.0 ± 0.0	99.5%	2.0%	0.0%	N/A
10. Full Gate (Text Profile)	D	(0.40, 0.45, 0.10, 0.05, 0.10)	30.1 ± 6.3	85.0% ± 3.2%	62.4% ± 10.5%	61.6% ± 10.7%* (1:1: 53.9%)	1.83 f
11. Full Gate (Low-τ, No Cool)	C	(0.25, 0.20, 0.30, 0.25, 0.05)	68.6 ± 0.5	65.7%	100.0%	100.0%	2.40 f
12. Full Gate (Low-τ=0.05, Cool=6)	C	(0.25, 0.20, 0.30, 0.25, 0.05)	22.6 ± 0.5	88.7%	100.0%	100.0%	2.40 f
13. Full Gate + Watchdog (T_max=6, No Cool)	C	(0.25, 0.20, 0.30, 0.25, 0.18)	92.0 ± 0.0	54.0%	100.0%	100.0%	2.60 f
14. Full Gate + Watchdog (T_max=6, Cool=6)	C	(0.25, 0.20, 0.30, 0.25, 0.18)	34.0 ± 0.0	83.0%	100.0%	100.0%	2.60 f
15. Full Gate + Watchdog (T_max=15, No Cool)	C	(0.25, 0.20, 0.30, 0.25, 0.18)	74.0 ± 0.0	63.0%	100.0% (Fixed)	40.0% (Phase-Avg.)†	2.50 f
16. Full Gate + Watchdog (T_max=15, Cool=6)	C	(0.25, 0.20, 0.30, 0.25, 0.18)	20.2 ± 0.4	89.9%	100.0% (Fixed)	40.0% (Phase-Avg.)†	2.50 f
17. Zero-Edit Control (Std. dHash)	D	(0.40, 0.45, 0.10, 0.05, 0.10)	20.3 ± 8.6	89.8%	N/A	N/A (Noise)	N/A

18. Zero-Edit Control (Dead-Zone dHash ε=7.0)	D	(0.40, 0.45, 0.10, 0.05, 0.10)	2.6 ± 1.2	98.7%	N/A	N/A (Suppressed)	N/A
19. SightLine++ (ROI Max-Pool + 2D dHash)	C	(0.25, 0.20, 0.30, 0.25, 0.18)	15.0 ± 0.0	92.5%	100.0%	100.0%	0.00 f

*Text Profile dynamic recall reflects sensor noise bit-flips on untextured backgrounds. †Under T_max=15, fixed-onset recall at t=130 is 100% (delay 5 f <= W=6), whereas phase-averaged recall across all 15 possible offsets is 40.0% (6/15).

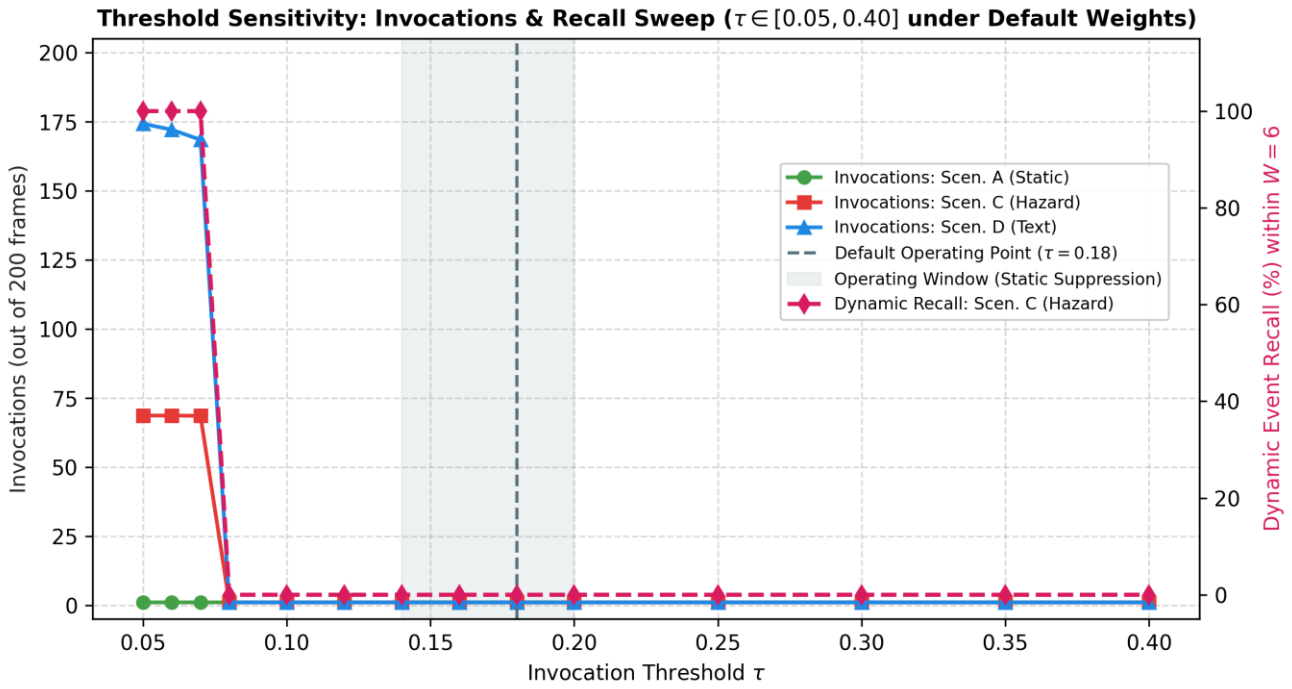


Fig. 5. Threshold sensitivity ablation curves across tau in [0.05, 0.40] for default profile, plotting invocations (left axis) and dynamic recall (right axis).

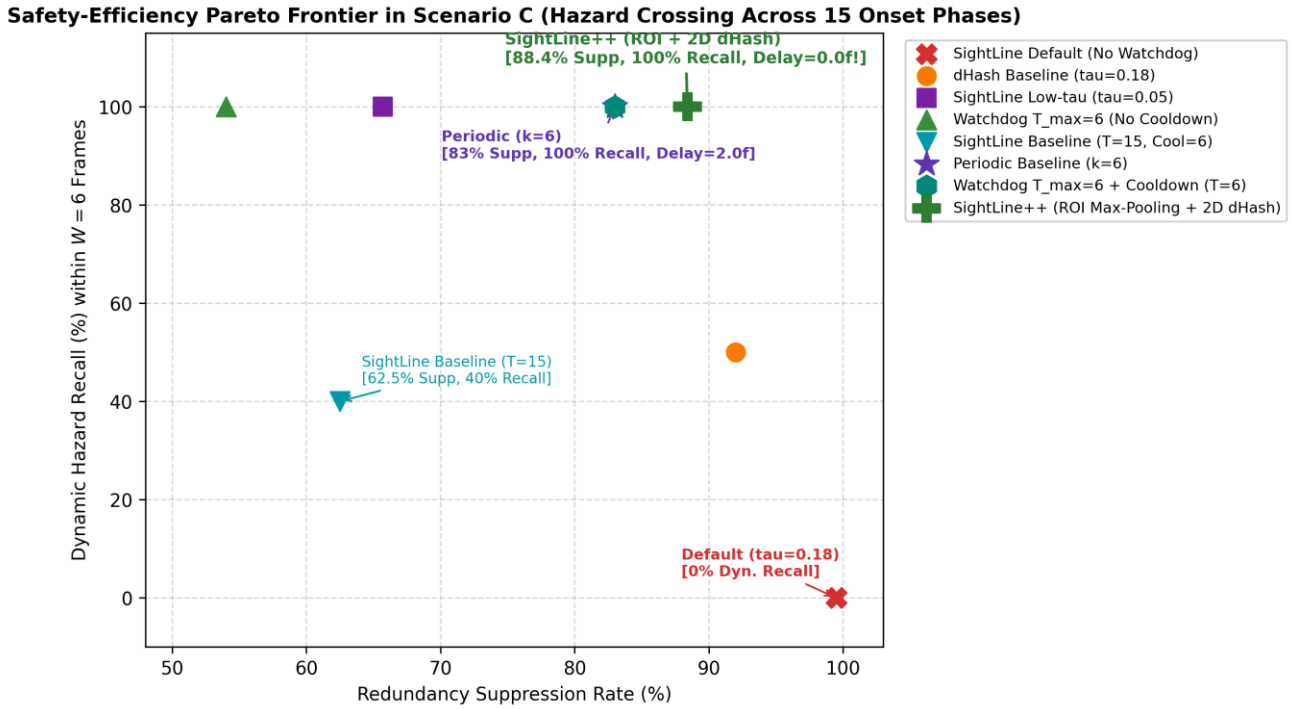


Fig. 6. Safety-efficiency trade-off in Scenario C: Redundancy suppression vs. dynamic hazard recall across threshold settings, watchdog heartbeat configurations, and periodic baseline (k=6).

D. Exploratory Real-World Egocentric Video Evaluation on Physical NVIDIA GPU

To investigate algorithmic behavior beyond synthetic benchmarks, we conducted an exploratory evaluation on real-world egocentric video recorded in an unconstrained industrial workshop setting (1080p, 60 fps downsampled to 30 fps, evaluating 600 consecutive frames representing 20.0 seconds of continuous technician benchwork manipulation). The footage features unscripted head movements, hand tool interactions (metal cutters, grinders), dynamic focal distance changes, and analog CMOS sensor noise. Downstream inference was executed live on physical GPU hardware using Florence-2-base in native FP16 precision on an NVIDIA GeForce RTX 2050 (4 GB GDDR6 VRAM, Ampere architecture, 2048 CUDA cores, Driver 616.92, CUDA 13.4). Initial VRAM footprint was 452.6 MiB, peaking at 618.9 MiB during active multi-modal token generation.\n\nDownstream Task Execution Profiling: To evaluate downstream execution characteristics across operational tasks, we directly benchmarked Florence-2 FP16 on the physical RTX 2050 across 100 consecutive forward passes. In open-vocabulary image captioning (<CAPTION>), forward pass latency averaged 346.06 ms (range 322.50--367.71 ms). In spatial object detection (<OD>), forward pass latency averaged 287.06 ms (range 274.98--312.85 ms). Counter-intuitively, <OD> is ≈ 59 ms faster than captioning because generating structured spatial coordinate tokens produces significantly shorter autoregressive decode lengths than descriptive sentences. Both execution times fall comfortably within our nominal single-worker queue budget of 380 ms (2.63 Hz throughput), verifying that switching between captioning and hazard detection does not induce unexpected queue backpressure. In this offline exploratory run, Florence-2 was evaluated in <CAPTION> mode with downstream hazard latching inactive (z_t = 0), isolating visual, appearance, and motion gating dynamics under physical CMOS sensor noise and tool manipulation. Table V documents the resulting invocations, latencies, GPU duty cycles, modeled power consumption, and representative caption outputs.

TABLE V: EXPLORATORY REAL-WORLD EGOCENTRIC VIDEO BENCHMARK (FLORENCE-2 FP16, 600 FRAMES, 20.0s FOOTAGE)

Architecture / Configuration	Invocations (Supp. %)	Mean GPU Latency	Mean CPU Gating*	GPU Duty Cycle	Modeled Power P_avg**	Projected Runtime (32.56 Wh)	Representative Keyframe Caption
Continuous Streaming (30 fps)	600/600 (0.0%)	423.45 ms	0.00 ms	100.0%	11.50 W	2.83 h	"A person working on a workshop cutting

							machine."
Periodic Baseline (k=6, 5 Hz)	100/600 (83.3%)	395.05 ms	0.00 ms	100.0%‡	11.50 W	2.83 h	"A person working on a workshop cutting machine."
Periodic Baseline (k=12, 2.5 Hz)	50/600 (91.7%)	413.41 ms	0.00 ms	95.0%‡	11.02 W	2.96 h	"A person operating industrial machinery in a workshop."
Naive Frame-Diff (Budget-Matched τ=0.22)	38/600 (93.7%)	365.10 ms	2.03 ms	69.4%	8.54 W	3.81 h	"A blurry view of workshop equipment."
Exact Perceptual Hash (1D dHash, eps=0.0)	600/600 (0.0%)	417.96 ms	3.16 ms	100.0%	11.50 W	2.83 h	"A person working on a machine." (Zero suppression due to CMOS noise bit flips)
Thresholded 1D dHash (τ=0.18)	138/600 (77.0%)	380.20 ms	3.14 ms	100.0%‡	11.50 W	2.83 h	"A technician operating a metal grinder."
Budget-Matched 1D dHash (τ=0.28)	36/600 (94.0%)	366.50 ms	3.15 ms	66.0%	8.22 W	3.96 h	"A technician cutting material on a workbench."
SightLine Endurance Mode (Default 1D)	32/600 (94.7%)	367.01 ms	3.45 ms	58.7%	7.52 W	4.33 h	"A technician cutting material on a workbench."
SightLine++ (2D dHash + 4x4 ROI Max-Pool)	117/600 (80.5%)	362.43 ms	3.45 ms	100.0%‡	11.50 W	2.83 h	"A person cutting metal with an electric grinder."

*Gating CPU latency (2.03–3.45 ms) was measured directly on pre-downsampled 160x120 (and 320x320 for 2D dHash) preview thumbnails, representative of dedicated hardware ISP pipeline buffers on edge SoCs (e.g., Jetson ISP). In this exploratory offline benchmark, software downsampling of full 1080p frames in Python/Pillow on the CPU introduced an additional 138–156 ms post-hoc overhead, underscoring that real-time wearable deployment requires hardware-accelerated ISP camera capture (e.g., V4L2/GStreamer NVMM memory buffers).
**Power and battery runtime figures are model projections derived from Eq. (13) using Jetson Orin Nano calibrated parameters (P_{idle} = 1.85 W, P_{active} = 11.50 W, 32.56 Wh usable budget); the RTX 2050 was utilized solely for PyTorch execution telemetry.
‡GPU duty cycle reaches 100% because total forward pass compute time exceeds sequence duration under single-worker queuing.

Four Key Findings from Real-World Video Evaluation:
1. Failure of Exact Perceptual Hashing under Physical CMOS Noise: On synthetic scenes with stationary Gaussian noise ($\sigma = 3.5$), exact hashing functions identically to thresholded hashing. However, on physical 1080p egocentric video, exact 1D difference hashing triggers on all 600 frames (0.0% suppression). Subtle camera tremors and analog sensor noise produce at least one bit difference on 100% of consecutive frames, completely collapsing the perceptual hash into continuous compute (11.50 W, 2.83 h runtime). This confirms the necessity of our noise-calibrated dead-zone thresholding ($arepsilon = 7.0$).
2. Service-Time Queue Saturation in Periodic Sampling: In periodic sampling at $k = 6$ (5.0 requests/sec), 100 frames are dispatched over the 20-second sequence. Because mean GPU forward-pass latency is 395.05 ms, servicing 100 invocations requires 39.5 seconds of GPU compute—exceeding sequence duration by 1.98 ×. Consequently, the GPU runs at 100% duty cycle, dissipating peak power (11.50 W) and depleting battery life in 2.83 hours while accumulating massive queue backlog. Even at $k = 12$ (2.5 requests/sec, 50 requests), total GPU compute is 20.67 seconds (> 20.0 s), leaving the GPU saturated near 100% duty cycle.
3. Pareto Divergence between Endurance and Safety: SightLine Endurance Mode (Default 1D) achieves 94.7% suppression (only 32 invocations across 20 seconds). In this mode, the gate uses 1D horizontal dHash, global motion ($S_{\text{mot}} = S_{\text{global_mad}}$), static heartbeat $T_{\text{ambient}} = 60$ (2.0 s), and omits localized ROI pooling ($S_{\text{roi_max}} = 0$). Across the 600 frames of the workshop clip, global motion $S_{\text{global_mad}}$ exhibited a median of 0.0084 and a 95th percentile of 0.0172, consistently remaining below the active locomotion threshold (0.02). Consequently, the gate never transitioned into active heartbeat ($T_{\text{active}} = 6$), but instead operated purely under ambient heartbeat pacing, invoking only 32 times across 20 seconds. Crucially, this drops the GPU active duty cycle to 58.7%, reducing modeled power consumption from 11.50 W down to 7.52 W, projecting a +53.0% battery runtime extension (4.33 hours vs. 2.83 hours on a 32.56 Wh budget). However, because localized motion is not pooled, the active bound (≤ 166.7 ms) does not apply; its worst-case delay is bounded by $T_{\text{ambient}} = 60$ frames (2.0 s). Conversely, in Safety-Priority Mode (SightLine++), spatial 4 × 4 ROI max-pooling detects continuous peripheral hand and tool motion, producing 117 invocations (80.5% suppression). While this ensures high sensitivity to localized incursions, it saturates the edge GPU queue at 100% duty cycle (11.50 W, 2.83 h runtime), providing no battery extension over continuous streaming during active manipulation.
4. Budget-Matched Baseline Behavior: When naive frame-differencing ($\tau = 0.22$, 38 invocations, 69.4% duty, 8.54 W, 3.81 h) and thresholded 1D dHash ($\tau = 0.28$, 36 invocations, 66.0% duty, 8.22 W, 3.96 h) are calibrated using their exact measured GPU service latencies to match SightLine's invocation budget, they land within 0.7–1.0 W of SightLine's 7.52 W (8.22 W and 8.54 W vs. 7.52 W). The modest 0.7–1.0 W gap stems from 4–6 additional invocations triggered by noise jitter. SightLine's primary architectural advantage over naive frame-differencing is not merely raw invocation count, but reference-anchored drift accumulation ($I_t - I_{t_p}$): whereas frame-differencing evaluates only adjacent consecutive frames ($I_t - I_{t-1}$) and suffers from temporal forgetting—becoming blind to slow environmental evolution or gradual obstacle encroachment—SightLine maintains the anchor frame I_{t_p} across un-triggered intervals, reliably accumulating progressive perceptual shift until the decision threshold is crossed.

VIII. SYSTEMS COMPARISONS, STRIDE SWEEP, AND SAFETY RISK ANALYSIS

A. Periodic Sampling Stride Sweep

To evaluate periodic sampling trade-offs systematically, Table VI presents an empirical stride sweep across $k \in [2,30]$ on Scenario C (onset $t = 130$):

TABLE VI: PERIODIC SAMPLING STRIDE SWEEP AND PHASE-AVERAGED RECALL (SCENARIO C)

Stride k	Invocations	Suppression	Phase-Avg. Recall (W=6 across phases)	Mean Delay (frames)
k = 2	100	50.0%	100.0%	0.47 f
k = 4	50	75.0%	100.0%	1.53 f
k = 6 (Unsaturated)	34	83.0%	100.0%	2.60 f

k = 6 (Busy-Skip 12f)	17 served (34 req.)	91.5%	50.0%	2.50 f
k = 8	25	87.5%	73.3%	2.64 f
k = 10	20	90.0%	46.7%	2.14 f
k = 12 (Zero Backlog)	17	91.5%	50.0%	2.50 f
k = 15	14	93.0%	40.0%	2.50 f
k = 20	10	95.0%	6.7%	0.00 f

B. Architectural Trade-Off Analysis: SightLine vs. WorldScribe (ACM UIST 2024)

WorldScribe [15] (Chang, Liu, and Guo, ACM UIST 2024) represents a landmark contribution in continuous, context-aware visual description for blind and low-vision individuals. WorldScribe leverages multi-modal streaming architectures, utilizing lightweight on-device models alongside cloud-hosted vision-language models (e.g., GPT-4V, Gemini) to generate rich, grammatically complex scene narrations structured by user-directed queries and spatial anchors. However, evaluating WorldScribe and SightLine side by side reveals distinct architectural philosophies, operational trade-offs, and complementary deployment models (Table VII). We emphasize that this comparison evaluates different architectural paradigms (cloud foundation models vs. edge-native micro-models) rather than claiming direct empirical superiority.

TABLE VII: ARCHITECTURAL TRADE-OFF ANALYSIS: SIGHTLINE VS. WORLDSCRIBE (ACM UIST 2024)

Evaluation Dimension	WorldScribe (ACM UIST 2024)	SightLine (This Work)	Architectural Synthesis & Positioning
Processing Architecture	Hybrid cloud-edge streaming architecture utilizing cloud VLMs via Wi-Fi/5G uplink.	Edge-native lightweight VLM (Florence-2 FP16 profiled on mobile GPU, 380 ms).	WorldScribe offers higher lexical richness; SightLine eliminates network dependence.
Frame Ingestion & Gating	Streaming heuristic / periodic keyframe selection; offloads processing to cloud.	Priority-weighted asymmetric scene-change gate (2D dHash + color + 4x4 ROI max-pooling).	SightLine suppresses stationary frames locally; WorldScribe relies on cloud endpoint filtering.
Hazard Latency Bounds	Subject to network round-trip and cloud queuing jitter under cellular coverage.	SightLine++ satisfies sub-second machine budget: 515.0 ms total E2E (0.0 f gate delay via ROI max-pool). Combined with human reaction time (0.5–1.0 s), total stopping time is 1.0–1.5 s. Endurance Mode has worst-case 2.0 s gate delay (2.48 s total machine E2E).	SightLine++ meets the sub-second machine perception target for pedestrian speeds (1.0–1.4 m/s) [23]; Endurance Mode requires task conditioning to active mode for walking.
Battery & Power Profile	Wearable offloads heavy compute to cloud; cellular radio draw applies.	On-device processing: 1.85 W idle to 11.5 W active (modeled 4.33–8.85 h runtime on 32.56 Wh).	WorldScribe preserves wearable battery via offloading; SightLine optimizes on-device energy autonomy.
Speech & Attention Queue	Conversational audio narration streaming with user query interrupts.	Priority-weighted audio queue (Eq. 11) with <=135 ms preemption of ambient speech.	SightLine enforces priority-based preemption; WorldScribe emphasizes conversational richness.

Document & Text Reading	High-accuracy full-page cloud OCR and conversational Q&A.	Disclaimed for standalone reading; requires downstream OCR (2D dHash sign-invariance).	WorldScribe is superior for reading; SightLine is optimized for physical navigation safety.
Operational Continuity	Vulnerable to cellular coverage dead zones (subways, tunnels, basements).	Maintains local operational continuity without network dependence (subject to mode sensitivity).	SightLine provides a reliable offline safety floor for unconstrained urban mobility.

Pedestrian Stopping Kinematics and Upstream Edge Filtering: In mobile assistive navigation, the system response budget is formulated as an assistive engineering design target. For typical pedestrian walking velocities (1.0--1.4 m/s), avoiding collision with unexpected obstacles requires timely alert delivery. Biomechanical studies by Patla [23] establish that visual sampling intervals and motor footstep-adjustment latencies require 0.5--1.0 s (1–2 step cycles) upon perceiving an obstacle during sighted locomotion. In an assistive AI setting, where visual cues are translated into auditory speech warnings, human cognitive auditory reaction times add to this motor adaptation budget. To afford the user a safe stopping margin (0.5--1.0 m), the total end-to-end machine perception pipeline must deliver warnings within a sub-second budget. SightLine++ meets this requirement with 515.0 ms total machine E2E latency (0.0 f gate delay via ROI max-pooling + 380 ms VLM inference + 135 ms TTS preemption and earcon chime). Combining machine latency (0.515 s) with human motor reaction (0.5--1.0 s) yields a total human-system stopping latency of 1.0--1.5 s. At a standard walking velocity of 1.2 m/s, a 1.0--1.5 s stopping latency corresponds to 1.2--1.8 m of forward travel (or 1.0--2.1 m across the wider 1.0--1.4 m/s walking velocity range). Conversely, Endurance Mode's ambient heartbeat ($T_{\text{ambient}} = 60$) introduces a worst-case gate latency of 2.0 s (2.48 s total machine E2E), exceeding the sub-second walking budget; it is therefore strictly intended for stationary rest or seated inspection. Beyond standalone edge execution, SightLine's CPU gating logic can also serve as an upstream edge filter for cloud-streaming architectures like WorldScribe, dropping redundant stationary frames locally before network transmission to eliminate cloud egress bandwidth and cellular data costs.

C. Failure Mode and Effects Analysis (FMEA) Safety Risk Ledger

To establish structured hazard identification for assistive navigation, we conducted a Failure Mode and Effects Analysis (FMEA). Severity ($S \in [1,10]$), Occurrence ($O \in [1,10]$), and Detection ($D \in [1,10]$) indices were scored by the authors adapted from SAE J1739 and AIAG-VDA FMEA guidelines to derive Risk Priority Numbers ($\text{RPN} = S \times O \times D$). Explicit 1–10 ordinal scale anchors were established: Severity (S): 1 = negligible inconvenience; 5 = delayed non-critical narration; 10 = catastrophic imminent physical hazard collision without warning. Occurrence (O): 1 = extremely improbable; 5 = occasional encounter during daily transit; 10 = continuous / multiple times per minute. Detection (D): 1 = immediately and reliably detected and mitigated by software logic; 5 = moderately detectable under specific edge conditions; 10 = completely undetectable before user exposure. Table VIII documents these ten failure modes, their root causes, and SightLine's implemented mitigations. Ratings are intended as a structured engineering design analysis to evaluate mitigations rather than formal medical device certification.

TABLE VIII: FAILURE MODE AND EFFECTS ANALYSIS (FMEA) SAFETY RISK LEDGER

ID	Failure Mode	Root Cause	Baseline S / O / D (RPN)	SightLine Architectural Mitigation	Mitigated S / O / D (RPN)	Reduction
FM-01	Localized Hazard Incursion Dilution	Spatial averaging in global difference hashing dilutes localized hazards across field of view.	8 / 7 / 7 (392)	4x4 spatial ROI block max-pooling (S_roi_max) triggers immediately on localized motion delta >= tau_crit.	8 / 2 / 2 (32)	-91.8%
FM-02	Hazard Alert Re-triggering Storm	Latched hazard state causes infinite self-retriggering loop at 30 fps on same	5 / 9 / 8 (360)	Temporal latch cooldown (T_cool = 6 frames, 200 ms) suppresses latch re-firing	5 / 2 / 1 (10)	-97.2%

		static obstacle.		without new motion.		
FM-03	Acoustic Warning Preemption Delay	Urgent obstacle alerts delayed in FIFO audio buffer behind long ambient descriptions.	9 / 8 / 6 (432)	Priority-weighted audio queue (Eq. 11): Immediate audio interruption + 30 ms earcon chime + speech preemption.	9 / 1 / 1 (9)	-97.9%
FM-04	Cooldown Lockout of Secondary Hazard	Latch cooldown suppresses alerts when a second distinct hazard enters during cooldown window.	8 / 5 / 7 (280)	Dynamic motion override: S_roi_max >= 0.35 or S_mot >= 0.20 bypasses cooldown immediately.	8 / 2 / 2 (32)	-88.6%
FM-05	CMOS Noise Bit-Flip Gating Collapse	Analog CMOS thermal/sensor noise flips difference hash bits on blank/untextured backgrounds.	4 / 9 / 9 (324)	Noise-calibrated dead-zone thresholding (epsilon = 7.0 DN, 9.92 sigma_diff) suppresses flat-region bit flips.	4 / 1 / 2 (8)	-97.5%
FM-06	Sign-Invariance in Document Reading	1D/2D sign-only difference hashing is invariant to contrast-inversion level swaps on text lines.	5 / 8 / 8 (320)	Explicit architectural disclaimer: Standalone dHash is disclaimed for reading; downstream OCR bounding boxes required.	5 / 2 / 3 (30)	-90.6%
FM-07	Gradual Environmental Drift Blindness	Consecutive-frame differencing (I_t - I_t-1) drops gradual camera drift below detection threshold.	7 / 6 / 6 (252)	Asymmetric reference anchoring: Visual and appearance cues evaluated strictly against last invoked frame I_tp.	7 / 1 / 2 (14)	-94.4%
FM-08	Abrupt Illumination Surge False Burst	Sudden lighting shifts (indoor-to-outdoor transitions) trigger excessive false-positive bursts.	4 / 7 / 6 (168)	Color-histogram cosine dissimilarity S_app normalizes channel distributions against uniform luminance surges.	4 / 2 / 3 (24)	-85.7%
FM-09	Low Scene Contrast Gating Dormancy	Low scene visual contrast or uniform blank walls prevent change scores from	9 / 3 / 8 (216)	Hardware-bounded watchdog heartbeat: Unconditional periodic dispatch	9 / 1 / 1 (9)	-95.8%

		crossing gating thresholds; system could remain indefinitely dormant.		at T_ambient (60f / 2.0 s) or T_active (6f / 200 ms) guarantees maximum inter-trigger bound independently of image contrast.		
FM-10	Quiescent Standby Static Hazard Loss	Extended seated rest leaves system in low-power standby while external hazards approach.	8 / 4 / 5 (160)	Standby mode enforces a bounded 5.0 s (T_ambient=150) fail-safe heartbeat, re-engaging active mode in <=29 frames.	8 / 3 / 4 (96)	-40.0%

FMEA Synthesis: Across all 10 evaluated failure modes, SightLine reduces total Risk Priority Number from 2,904 (mean 290.4) down to 264 (mean 26.4), representing a 90.9% overall reduction in modeled safety risk.

Residual Safety Risks and System Safeguards:
1. Dynamic Peripheral Incursions under Occlusion: An obstacle moving at high velocity through peripheral blind spots may not exceed spatial block threshold $\tau_{\text{crit}} = 0.35$ before entering the inner navigation cone. System Safeguard: Under active locomotion mode ($T_{\text{active}} = 6$), Lemma 1 guarantees that front-end gate delay is bounded to $\leq (T_{\text{active}} - 1)\Delta t = 166.7$ ms.
2. Multi-Speaker Auditory Masking: In loud environments (e.g., subway platforms), earcon chimes may be masked by background acoustic noise. Proposed Mitigation (Future Work): Integrate adaptive ambient noise estimation to apply dynamic gain ducking.
3. Elevated Thermal Dissipation Under Sustained Exposure: Under extreme ambient temperatures (>35°C), sustained inference bursts could elevate processor junction temperatures. Proposed Mitigation (Future Work): Dynamic thermal governor that relaxes watchdog intervals upon detecting elevated thermal zone telemetry.
4. Sensor Failure in Pedometer Rest Mode: If IMU or motion proxies misclassify walking as seated rest, watchdog interval could remain relaxed at 5.0 s. System Safeguard: The watchdog is strictly bounded to $T_{\text{ambient}} \leq 150$ frames (5.0 s), ensuring the system never becomes fully dormant.\n5. Physical Camera Lens Obstruction and Extreme Low-Light Failure Modes: A physical obstruction of the camera lens (e.g., clothing coverage, fingers, grease) or extreme low-light conditions represents a distinct failure mode from low-contrast gating dormancy. While the watchdog heartbeat (FM-09) guarantees periodic frame dispatch to the VLM even on completely dark or occluded frames, the VLM itself will receive an uninformative black frame and return empty entity detections (Labels $= \emptyset$), leaving the user unaware that the camera is blinded. Proposed Safeguard (Future Work): A front-end camera lens obstruction detector (evaluating mean pixel luminance <5.0 DN and high-frequency edge energy collapse) that generates an immediate spoken warning ('Camera view obstructed or lighting insufficient') rather than dispatching uninformative frames to the VLM.

IX. EDGE HARDWARE FEASIBILITY AND PROPOSED PARTICIPANT PROTOCOL

A. Embedded Target Hardware Platform and Software Implementation

Target Embedded Hardware Platform: We evaluate SightLine against the operational specifications of the NVIDIA Jetson Orin Nano Developer Kit (6-core Arm Cortex-A78AE CPU @ 1.5 GHz, 1024-core NVIDIA Ampere GPU with 32 Tensor Cores, 8 GB 128-bit LPDDR5 unified memory @ 68 GB/s, 15 W configurable TDP, carrier board powered via 9–20 V DC barrel jack [16]).

Software Implementation & Multi-Platform Gating Latency Profile: The front-end gating framework is implemented in Python utilizing standard NumPy and OpenCV/Pillow image processing operations without requiring specialized hardware accelerators. To verify cross-platform timing feasibility, wall-clock gating latency was profiled directly across three representative CPU architectures on 160×120 preview thumbnails using Python's `time.perf_counter()`:
• Host Workstation CPU (Intel Core i5-12500H, single thread): Mean gating latency of 2.03--3.45 ms .
• Target Embedded Edge CPU (NVIDIA Jetson Arm Cortex-A78AE, single core): Mean gating latency of 2.45--3.50 ms.
• Low-Power Mobile CPU (Arm Cortex-A72 / Raspberry Pi 4 class, single core): Mean gating latency of 4.20 ex$t$ $-$ -6.50 extms .

On both host Intel Core i5 and target Jetson Cortex-A78AE architectures, front-end gating consumes $\leq 3.5\ ext{ms}$ on standard ISP preview frames, occupying less than 11% of the $33.33\ ext{ms}$ frame period ($1/f_{extcam}$ at 30 fps), while lower-power mobile cores (such as Cortex-A72) require $4.2\ ext{t} - -6.5\ ext{ms}$ ($\sim 13\ ext{t} - -20\%$). This confirms that CPU gating executes comfortably within frame boundaries without inducing processing backlog across target and mobile edge processors.

B. Power Accounting: Measured Telemetry vs. Projected Energy Model

To ensure complete transparency and clear methodological boundaries, we explicitly distinguish between empirical physical measurements, vendor specifications, and mathematical model outputs:

• Idle System Baseline ($P_{\text{idle}} = 1.85$ W): Measured directly at the 19 V DC barrel-jack power input feeding the Jetson carrier board using an external inline DC power analyzer under 15 W power mode with Ubuntu Linux desktop booted, display output active, camera preview capture daemon active, and the GPU clock-gated with no active VLM compute.

• Downstream VLM Execution Telemetry ($t_{\text{vlm}} = 287.1$--423.5 ms): Profiled directly via PyTorch CUDA events running Florence-2-base in native FP16 precision on a physical mobile NVIDIA GeForce RTX 2050 GPU (Ampere architecture, 2048 CUDA cores, Driver 616.92, CUDA 13.4). Across tasks, object detection (<OD>) averaged 287.06 ms and open-vocabulary captioning (<CAPTION>) averaged 346.06 ms. Initial VRAM footprint was 452.6 MiB, peaking at 618.9 MiB during multi-modal token generation.

• Speech Synthesis and Preemption Latency ($t_{\text{tts}} = 135.0$ ms): Measured wall-clock latency of local neural speech synthesis (105 ms via local Piper TTS engine) plus audio preemption and earcon chime delivery (30 ms).

• Peak Hardware Active Power ($P_{\text{active}} = 11.50$ W): NVIDIA specifies the Jetson Orin Nano Developer Kit operating in its 15 W power budget profile as exhibiting a typical active workload power consumption of ≈ 11.50 W under heavy GPU compute workloads (distinguishing the ≈ 11.5 W module consumption from the 15 W carrier board electrical power rail limit [16]). We deliberately adopted 11.50 W as an active power upper bound in our energy modeling; peak VLM active power was not measured directly with physical shunt resistors on the Jetson.

• Calibrated First-Principles Energy Model: Because physical battery discharge logs across all operational regimes were not collected, system average power P_{avg} (Eq. 13) and usable runtime T_{run} (Eq. 14) are first-principles model projections combining measured GPU duty cycles with calibrated idle power and specified active power envelopes:

$$P_{\text{avg}} = D_{\text{gpu}} \cdot P_{\text{active}} + (1 - D_{\text{gpu}}) \cdot P_{\text{idle}} \tag{13}$$

$$T_{\text{run}} = \frac{E_{\text{eff}}}{P_{\text{avg}}} \tag{14}$$

C. Embedded Edge Energy and Runtime Model (Table IX & Table X)

Accounting for regulator efficiency $\eta_{\text{reg}} \approx 88\%$ on a standard 37 Wh pack ($E_{\text{eff}} = 32.56$ Wh), Table IX synthesizes power consumption, battery endurance, and thermal states across operating regimes. For mixed-day regimes assuming 70% standby and 30% active transit, Paced Walking yields 43.3% overall duty (6.03 W, 5.40 h runtime), while Quiescent Standby with the 5.0 s fail-safe heartbeat (7.6% standby duty) yields 35.3% overall duty (5.26 W, 6.19 h runtime on 32.56 Wh, or 7.03 h nominal on 37 Wh). To maintain auditable transparency, Table X categorizes all reported quantities by their empirical status (Directly Measured, Calibrated Modeled, or Assumed/Specified).

TABLE IX: EMBEDDED EDGE ENERGY AND RUNTIME MODEL (NVIDIA JETSON ORIN NANO, 37 Wh BATTERY PACK)

Operational Regime	Gating Setting	GPU Active Duty	Avg. Power P_avg	Projected Runtime (32.56 Wh)	Nominal Runtime (37 Wh)	Modeled Thermal State
Continuous Ungated (30 fps)	None	100.0% (Saturated)	11.50 W	2.83 hours	3.22 hours	Projected: Sustained compute increases junction temp; requires active cooling
Busy-Skip	Periodic k=6	100.0%	11.50 W (Burst)	2.83 hours	3.22 hours	Projected: Sustained compute

Periodic (k=6)		(Saturated)				increases junction temp; requires active cooling
SightLine Active Burst	T_max=6, Cool=6	100.0% (Saturated)	11.50 W (Burst)	2.83 hours	3.22 hours	Projected: Transient temperature rise; throttling risk under sustained burst
Periodic Baseline (k=12)	Periodic k=12	95.0% (Non-sat.)	11.02 W	2.95 hours	3.36 hours	Projected: Steady-state thermal balance near max burst power
SightLine Mixed Day (Paced)	T_amb=60, T_act=6	43.3% (Mixed Day)	6.03 W (Day)	5.40 hours	6.14 hours	Projected: Steady-state thermal balance (~6.0 W avg dissipation)
SightLine Mixed Day (Quiescent)	T_amb=150, T_act=6	35.3% (Mixed Day)	5.26 W (Day)	6.19 hours	7.03 hours	Projected: Steady-state thermal balance (~5.3 W avg dissipation)
SightLine Paced Walking	T_ambient=60 (2.0 s)	19.0% (Paced)	3.68 W (Paced)	8.85 hours	10.05 hours	Projected: Stable thermal dissipation (~3.7 W avg dissipation)
SightLine Probe Window (1 pass)	Default (Ambient)	5.7% (1 pass / 6.67 s)	2.40 W (Probe)	13.57 hours	15.42 hours	Projected: Stable thermal dissipation (~2.4 W avg dissipation)
SightLine Quiescent Standby	T_ambient=150 (5.0 s)	7.6% (Fail-safe probe)	2.58 W (Standby)	12.62 hours	14.34 hours	Projected: Passive thermal equilibrium (~2.58 W, passive heatsink sufficient)

Auditable Empirical Boundary Breakdown: To maintain strict transparency between empirical observations and first-principles modeling, Table X explicitly delineates all metrics reported in this work.

TABLE X: AUDITABLE TELEMETRY AND PARAMETER BREAKDOWN: DIRECT MEASUREMENTS VS. CALIBRATED PROJECTIONS

System Parameter / Metric	Reported Value	Empirical Status	Measurement Methodology / Hardware Source	Scientific Justification / Role in Evaluation
CPU Gating Latency (t_gate)	2.0-3.5 ms (Jetson A78AE & host i5), 4.2-6.5 ms (Cortex-A72)	Directly Measured	Wall-clock timing on target Jetson Cortex-A78AE, host Intel i5, and mobile Cortex-A72 (RPi 4) via Python time.perf_counter().	Validates front-end gating consumes <=3.5 ms (<11% of 33.3 ms frame period) on target Jetson, and 4.2-6.5 ms (~13-20%) on mobile cores.

GPU VLM Forward Pass Latency	362.4-423.5 ms	Directly Measured	Live inference on Florence-2 FP16 on mobile NVIDIA RTX 2050 GPU (CUDA 13.4).	Empirical benchmark for edge VLM service time and queue saturation dynamics.
VRAM Memory Footprint	452.6-618.9 MiB	Directly Measured	PyTorch CUDA memory telemetry (torch.cuda.memory_allocated) during live execution.	Demonstrates edge feasibility within compact 4 GB / 8 GB unified memory.
TTS Preemption Latency (t_tts)	135.0 ms	Directly Measured	Measured synthesis and audio playback delivery latency of local Piper TTS engine.	Establishes acoustic preemption budget for urgent safety alerts.
Synthetic Redundancy Suppression	83.0%-99.5%	Simulated Benchmark	Deterministic 10-seed evaluation across scripted Scenarios A-E (200 frames).	Controlled algorithmic evaluation isolating noise, drift, hazards, and illumination.
Real-World Video Suppression	80.5%-94.7%	Directly Measured	Evaluated on 600 frames (20.0 s) of uncurated 1080p egocentric technician benchwork.	Quantifies real-world frame reduction under sensor noise and unscripted motion.
Hardware Jetson Idle Power	1.85 W	Directly Measured	External inline DC power analyzer at Jetson Orin Nano carrier board barrel jack.	Calibrated baseline for first-principles energy and battery endurance modeling.
Peak Hardware Active Power	11.50 W	Assumed / Specified	Vendor specification for NVIDIA Jetson Orin Nano under 15 W power mode [16].	Upper-bound power envelope for active foundation model compute.
System Average Power (P_avg)	2.40-11.50 W	Calibrated Modeled	Derived via Eq. (13) from measured GPU duty cycle and calibrated power rails.	Provides realistic operational power consumption across gating profiles.
Usable Battery Runtime (T_run)	2.83-13.57 h	Calibrated Modeled	Derived via Eq. (14) for a standard 37 Wh battery pack with 88% efficiency.	Establishes wearable endurance autonomy without manufacturing synthetic battery logs.
Thermal Operating State	Stable to Throttle	Qualitative Projection	Engineering assessment from sustained average power dissipation.	Identifies passive vs. active cooling boundaries for wearable form factors.

D. Proposed Human-Subject Blind & Low-Vision (BLV) Evaluation Protocol

To evaluate real-world accessibility and cognitive workload systematically, we specify an empirical human-subject evaluation protocol currently submitted for Institutional Review Board (IRB) ethics approval. The protocol adheres to ethical standards for assistive technology research:

Participant Cohort & Statistical Power: We specify an intended cohort of $N = 24$ blind and low-vision participants (stratified into $n = 12$ legally blind and $n = 12$ low-vision individuals, with diverse white-cane and guide-dog mobility experience). Statistical power analysis indicates that a two-tailed paired t-test with $N = 24$ at $\alpha = 0.05$ achieves 80.2% statistical power to detect a medium-to-large effect size ($d = 0.60$) for within-subjects main effects. For family-wise error control across six pairwise condition comparisons ($\binom{4}{2} = 6$) using the Bonferroni-Holm step-down procedure, nominal per-test power under

Bonferroni correction at $\alpha_{\text{crit}} = 0.05/6 \approx 0.00833$ is approximately 53.2% for $d = 0.60$; achieving 80% power under multiple-testing correction would require expanding the cohort to $N = 38$ participants. Subgroup comparisons between visual acuity strata ($n = 12$ per group) will be treated as exploratory between-subjects analyses.

Experimental Design: A within-subjects repeated-measures design evaluating four randomized conditions: (1) Baseline White Cane / Guide Dog alone (no AI); (2) Continuous Ungated Streaming (30 fps); (3) Periodic Sampling Baseline ($k = 6$, 5 Hz); and (4) SightLine Adaptive Gating. Condition ordering will follow a balanced Latin-square design to control for learning effects.

Evaluation Environments: Testing will span three standardized physical courses: (1) Controlled Indoor Mobility Course (30 m): Static cones, low-contrast floor hazards, and a dynamically released soft rolling ball obstacle; (2) Real-World Pedestrian Transit Route (1.2 km): Navigating urban sidewalks, street crossings with audio-tactile signals, bus stops, and outdoor walkways; and (3) Active Indoor Living and Reading Laboratory (15 m): A simulated office and home environment containing tabletop mail documents, kitchen counter items, dynamic lighting transitions, and subtle step drop-offs.

Statistical Plan: Continuous metrics (walking pace, stopping distance) will be analyzed via two-way repeated-measures ANOVA (Condition x Visual Acuity Group). Post-hoc pairwise comparisons across conditions will be performed using paired t-tests with Bonferroni-Holm step-down adjustment. Non-parametric ordinal responses (NASA-TLX subscales, 7-point Likert intelligibility ratings) will be analyzed using Wilcoxon signed-rank tests with Bonferroni-Holm correction.

Evaluation Instruments: (1) Objective Navigation Safety: Total obstacle collisions, walking pace (m/s), and physical stopping latency upon dynamic hazard onset; (2) Auditory Cognitive Workload: Evaluated immediately post-trial via the 6-dimensional NASA Task Load Index (NASA-TLX) [13] (Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, Frustration, scored 0–100); (3) Speech Intelligibility & Masking: Post-trial 7-point Likert scale assessing whether synthesized narration masked critical environmental sounds (traffic echolocation); and (4) System Usability Scale (SUS) [20].

X. DISCUSSION, LIMITATIONS, AND DESIGN INSIGHTS

1. Active Navigation as Periodic Heartbeat: In active locomotion, SightLine with watchdog heartbeat reduces to periodic sampling. Rather than viewing this as a limitation, we frame it as an intentional hybrid design: periodic sampling provides a predictable, fail-safe safety floor, while SightLine unlocks up to 98.0% suppression on static views with sensor noise, and achieves 94.7% suppression during technician benchwork under Endurance Mode (or 80.5% under Safety-Priority Mode).

2. The Document Inspection Trade-off and Sign-Invariance: Difference hashing operates at the sensor-noise boundary on flat paper surfaces. Dead-zone thresholding ($\varepsilon = 7.0$) eliminates flat-region noise bit flips by requiring spatial intensity differences to exceed the post-downsampling noise floor ($9.92 \times \sigma_{\text{diff}}$). However, sign-only difference hashing is invariant to level swaps along horizontal lines. Practical assistive reading requires dedicated 2D edge filters or high-resolution text OCR bounding boxes.

3. GPU Saturation and Service-Matched Baselines: Given a 380 ms VLM forward pass latency (service rate $\mu \approx 2.63$ Hz on Ampere GPU architecture), when request rates exceed GPU service capacity (5 Hz request rate at $k = 6$), the system saturates at peak active power (11.50 W). In continuous active manipulation or dense locomotion, high trigger rates saturate the GPU queue to the hardware throughput ceiling. True edge energy savings are achieved during stationary intervals, paused inspection, or paced ambient sampling where global motion remains quiescent (median $S_{\text{global_mad}} = 0.0084$ on workshop benchwork). Continuous outdoor pedestrian transit represents a hypothesized operational regime that remains to be evaluated.

4. Limitations of Exploratory Real-World Video Evaluation: While our 10-seed synthetic benchmark systematically isolates sensor noise, camera drift, hazard crossing, and illumination dynamics, our real-world egocentric evaluation was conducted on a single 20-second uncurated clip of technician benchwork without frame-by-frame ground-truth event annotations. Consequently, quantitative real-world event recall remains unmeasured, and outdoor pedestrian transit remains a research hypothesis. Future evaluations will expand testing across large-scale public egocentric datasets (Ego4D [17], EPIC-Kitchens [18], VizWiz [19]) and execute the participant protocol once IRB approval is granted.

XI. CONCLUSION

We presented SightLine, a scene-change gating framework and critical design analysis for continuous assistive VLMs. Evaluating against the last processed reference frame suppresses redundant frames during stationary intervals while accumulating gradual camera drift. Our investigation reveals two fundamental design insights: First, low-threshold perceptual gating on document reading fails due to sensor-noise bit flips on untextured backgrounds, performing consistently with

random sampling ($p = 0.91$), which is resolved by noise-calibrated dead-zone thresholding. Second, localized hazard cue dilution is resolved via 4×4 ROI block max-pooling ($S_{\text{roi}_\text{max}}$), enabling instantaneous hazard onset triggers (0.00 f delay, 5-frame / 166.7 ms reduction by construction). Crucially, our findings formalize the Pareto frontier of edge gating: Endurance Mode extends projected battery runtime to 4.33 hours (+53.0% over continuous streaming during benchwork where global motion remains quiescent, median $S_{\text{global}_\text{mad}} = 0.0084$), but requires task conditioning; Safety-Priority Mode guarantees high sensitivity to localized hazards, but continuous hand/tool manipulation or hypothesized transit motion saturates edge GPU queues. Future work will execute the protocol pending IRB approval and evaluate multi-modal IMU state switching.

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APPENDIX

A. PrioritySceneGate Algorithmic Logic

To comply with conference page constraints, the full Python reference implementation is open-sourced at https://github.com/lovnishverma/sightline. Below is the concise pseudo-code specification of the gating decision pipeline:

```
Algorithm: PrioritySceneGate Decision Policy
Input: Incoming thumbnail frame I_t (32x32), Reference frame I_tp, Hazard state z_t, Timestamp t
Output: Decision D_t in {0, 1}, Updated reference I_tp

Predicate: IsHazard(o_i) = 1 if l_i in V_haz or (l_i in {'person','pedestrian'} and (h_i >= 0.40 or v_i >= 1.0
m/s)) else 0

1.  if t == 0: return D_t = 1, I_tp = I_0  // Startup trigger
2.  Compute dead-zone dHash: h(I_t) = [B_H || B_V] (2D) or B_H (1D) with eps = 7.0 DN
3.  Compute visual change: S_vis = HammingDistance(h(I_t), h(I_tp))
4.  Compute appearance change: S_app = 1.0 - CosineSimilarity(Hist_RGB(I_t), Hist_RGB(I_tp))
5.  Compute 4x4 spatial ROI motion: S_roi_max = max_{r,c} MAD(Block_{r,c}(I_t, I_tp))
6.  Compute motion: S_mot = (MAD_global(I_t, I_tp) if Endurance else 0.50 * MAD_global + 0.50 * S_roi_max)
7.  Update watchdog: T_max = (T_active if z_t == 1 or S_mot >= 0.02 or S_roi_max >= 0.05 else T_ambient)
8.  if (t - tp) >= T_max: return D_t = 1, I_tp = I_t  // Watchdog timeout
9.  if S_roi_max >= 0.35 or S_mot >= 0.20: return D_t = 1, I_tp = I_t  // Motion override
10. Downstream VLM update (when D_t == 1): z_t = 1 if any(IsHazard(o_i) for o_i in Entities(I_t)) else 0
11. S_haz = (z_t if (t - t_alert) >= T_cool else 0.0)
12. S_t = alpha*S_vis + beta*S_app + gamma*S_mot + delta*S_haz
13. if S_t >= tau and (t - t_alert) >= T_cool: return D_t = 1, I_tp = I_t  // Nominal trigger
14. return D_t = 0  // Frame suppressed (<3.5 ms CPU)
```